

Public–Private Partnership (PPP) as a Social Game

PETER SCHARLE

ABSTRACT *Discrepancy between social demands and public resources has become apparent in many countries worldwide. Institutions and industries involved in public services seek new forms of collaboration and combine all available public and private skills to circumvent problems. Advocates of public–private partnership (PPP) assess the progress as convincing. Reserved observers argue that the technique has immanent weaknesses. Putting the phenomena, experience, and discussion into a wider context of gaming helps to arrive at a better comprehension. The paper focuses on road transportation infrastructure issues but its conclusions are of broader interest. Four perspectives are chosen to discuss PPP as a gaming issue. Linguistics, the mathematical theory of games, experimental psychology and the theory of institutions offer useful analytical concepts, theorems and approaches. The gaming perspective allows us to identify and define more clearly several uncertain or misinterpreted situations or actions experienced in PPP implementations. The conceptual identification of positions, motivations and further characteristics is not only possible but helps all partners to better understand the structure, the rules and roles given in the procedure. Use of the gaming perspective in PPP issues seems to enhance ‘fair play’. Expected trends of development and further areas of suggested research are derived from the conclusions and conjectures. Application of the gaming perspective is recommended in the development of project delivery techniques, education and training.*

Introduction

Partnerships between various parties committed to their particular interests but ready to co-operate for common goals have always been considered as trustworthy connections. The positive sense of the word has not been eroded yet and reflects some expected or immanent confidence. This fact might have influenced politicians as well as industrial and administrative leaders who coined the expression ‘public–private partnership’ (henceforth PPP) and started to experiment with various practical implementations of co-operation between the public and private sectors. In this connection it would be useful to choose new and neutral perspectives and reconsider the whole *game*. This way, the existing PPP literature can be augmented with considerations from other disciplines such as sociology, psychology or political science.

This paper presents the results of such an effort. While the perception of PPP and its practical applications is kept as broad as possible, the detailed analysis is restricted to transportation infrastructure development. The focus is on the accumulated experience and recent state of PPP applications in the USA, particularly with regard to transportation policy implementation. Much of the experience and lessons learned in the European PPP practice during the last two decades, well known and analysed elsewhere (Martinand, 1994; Turró, 1999), have not been mentioned or are referred to only briefly. It is

left to a forthcoming study to identify the common conclusions by comparing particular business-culture-dependent alternatives and trends.

Several compromises in the wording and in the argumentation have been made in favour of readers with some engineering background. It was one of the objectives of the research reported in this paper to enhance the comprehension of PPP issues for those having expertise in transportation and urban environment hardware. They are often on board without appropriate understanding of the game played. Thus, some of the conclusions serve an educational purpose.

The public-private partnership phenomenon

By definition, the public sector is expected to attend to community interest, stewardship and solidarity. The private sector is thought to be creative and dynamic, possessing capital, managerial efficiency, professional knowledge and entrepreneurial spirit. When the discrepancy between social demands and public resources becomes apparent, and neither decreasing the former nor increasing the latter seems to be feasible, it is a natural response to try to combine all available skills to circumvent problems. Governments at different levels, political parties, financing institutions and all the industries involved in public services or facilities of any sort (such as education, health care, utilities, mass transportation, etc.) seek new synergetic forms of collaboration to satisfy expectations and interests.

As a result of these efforts, the number of PPP implementations grew continuously during the past two decades (Appendix A presents several examples). It has been shown that the technique works almost everywhere, although essential differences exist between the existing conditions and challenges in the different areas of application. A vast literature emerged to report on principles and guidelines, as well as convincing successes and disappointing failures. Advocates of PPP assess the progress as slow but convincing. Others argue that the technique has immanent weaknesses, which limit the scope of successful application.

Some features are particularly suitable to distinguish the dispositions of the two sectors:

- Public administration is expected to be accountable and transparent. Furthermore, there are services that governments can do better than private business, or even that only governments can do.
- Private enterprises and, especially, financial institutions are resistant to disclosure of their moves with reference to the market value of the efficient or innovative business skill.

These differences emerge quite strongly in PPP projects. Overlooking, down-playing or pitting one against the other can encumber the clear understanding of the games played.

The gaming¹ perspective

Gaming can be defined as an activity involving more than one participant, where the following attributes are of characteristic importance:

- the moves of the participants correspond to a pattern of complementary roles;
- the participants (either individuals or groups) have specific goals and driving strategies;
- the moves are motivated by the outcome of the game and by the pay-offs.

In other words, where *rules, roles, outcomes, strategies and pay-offs* offer a framework to describe a procedure the game proves to be a useful concept of modelling. Both live

(real-time) and constructed (simulated) procedures can be perceived as games. In the case of real-time procedures, *identification* is a crucial issue. In constructed games, their structure and *usage* are of primary interest.

The identification of live games is a question of modelling. It is left to the observer to create a model of adequate approximation and to verify this by observations or experiments. Experience shows that important real-time procedures can be modelled adequately as games:

- military activities have been modelled and identified as games and their constructed variants are often used for training and experimental purposes;
- economic procedures and processes are described as games played with rules, and a comprehensive mathematical approach ('game theory') has been developed to solve complicated quantitative problems;
- psychological games played by people with concealed motivations and ulterior motives are classified and used in consulting rooms;
- politicians construct games when they choreograph legislative regulations or play them when bargaining votes in constitutional bodies;
- the theory of 'social capital' can be explained and applied using the basic concepts of the theory of games.

It is worth distinguishing between gaming and the mathematical 'theory of games'. Many of the concepts of game theory are of utmost importance in order to understand and play a game of any sort. However, a game is not simply a complicated (or complex) pattern of actions tractable with a rigorous mathematical theory. In many cases the gaming perspective turns out to be useful even if the basic assumptions of game theory (for instance, the players are rational utility-maximizing actors) are not fulfilled (Shubik, 1975). The language of game theory provides a way of organizing one's thoughts to identify, understand or design—and occasionally enjoy—a game.

There is a traditional interaction among the scholars involved in the development of particular gaming theories. Linguists gave detailed social analyses about the big 'con games' that flourished in the early 1900s (Maurer, 1940). Psychologists have been influenced by the virtuosity of the impostures played in business (Berne, 1964). Gaming concepts introduced in transactional psychology influenced economists (Shubik, 1975, 1982). Mathematicians have been involved in military (such as disarmament or deterrence) issues and political (e.g. voting or coalition-creating) applications (Shepsle, 1999).

A public–private partnership (PPP) in the transportation sector can be seen as that particular relationship between entities responsible for and interested in constructing, operating, maintaining and improving the transportation infrastructure. Evidence is available that there are rules, roles, strategies and pay-offs governing PPP procedures. It is worth analysing them from the perspective of game theories, as well as modelling them as complex games, in order to better understand the failures and difficulties.

Experience, lessons and trends of application

The co-operation between governmental and private entities to build and operate public services has a long tradition. The private ownership and operation of transportation facilities was common practice both in Europe and in the USA in earlier centuries (Albert, 1972; Lorrain, 1994; Giglio, 1997; Miller, 2000). Its importance and forms of implementation have developed according to the changing economic, social and political institutions of societies.

Papers and statistics presented by the promoters of PPP often have the air of advertisement; suggestive all-purpose arguments are illustrated with a few facts about the subject. Nevertheless, there is no lack of good studies that survey the activity, collect the experience and highlight the most important lessons (Ferrigno, 1990; Lockwood, 1995, 1997; Lockwood *et al.*, 2000). A much smaller group of authors discuss PPP with some reservation. While the importance of partnership and innovative financing are appreciated in their papers, these authors keep a distance, question the merits and discuss the weaknesses of the applications. Academic scholars analyse a few of the failures where facts and data are available and bring to light several questions ignored by supporters. Some of the papers belonging to this group extend the discussion to social issues as well. *Promoters and appliers* of PPP schemes often follow examples of commercialization taken from other sectors where direct-user financing, private management and turnkey project delivery are the norm. Good examples are available in the water supply and waste-water management industry as well as in the energy (power) sector. This fact would suggest extending the analysis to other industries. There are, however, two reasons why this paper focuses on the road sector:

- Significant technical differences follow from the diverse characteristics of the different public services. The measurability of use, the identification of the user as well as the possible costs of user-fee collection, all depend on whether the facility is 'fenced' or not and on whether end users subscribe in advance to the implementation of the facility. Even within the public services of the transportation sector there are considerable differences between the principles and techniques of charging users. Road-use charging seems to be one of the most sophisticated issues where solutions applied successfully elsewhere do not work.
- The social element in PPP schemes is less service-type related. People seem to follow rather general incentives, considerations and motivations. Charging techniques have to be suited more to the characteristics of the given facility than to the human behaviour of the prospective user being stereotypical. Private financing interests are seldom restricted to a single type of public service; revenue expectations prevail over almost any other consideration.

For the road transportation industry, demands are a function of economic growth, sub-urbanization, car ownership and disposable income. Traditional public incomes (motor fuel and highway-use-related taxes) fail to keep pace with the costs of operation, maintenance and development, because they are either too small or used for other purposes. In the early 1990s, for example, it was stated that the annual public resource of \$40 billion in the USA would not be sufficient to cope with the accruing physical preservation needs of the mature federal road network. It is commonly believed that better management and innovative financing are the right answer and that both are offered by inventive PPPs.

In response, in the past two decades direct and indirect beneficiaries of infrastructure improvements (such as property owners and developers) have been more explicitly identified and taxed according to the assessment of their benefits. In many countries legal procedures have been established to regularize cost-sharing techniques. Governmental authorities created new institutions with a varying degree of responsibility (e.g. transportation development corporations, road utility districts) to develop and finance public road infrastructure. Private profit-making entities (mixes of investment, design, construction, management and technology companies) have been searching new roles beyond design and construction; they hold concessions to finance, build, operate and possibly own a highway. The consortia collect toll revenues sufficient to repay project debts and

earn a reasonable return on investment. Several states and local governments impose economic regulation on them by reviewing the toll rate charged by the owner (Virginia), or by controlling the return on investment with toll rates left to the concessionaire (California).

National and supranational transportation policies have incorporated the basic principles and promote the application of innovative public–private financing. As a pioneering document in the USA, the ISTEA² permits:

- commingling of conventional fuel vehicle tax-derived funds with toll-backed revenues;
- the extension of federal aid to cover private or quasi-public entities;
- the creation of flexible funding mixes and financing instruments.

Legal and policy documents in the USA and in the EU as well as state-level regulations (such as the Public–Private Transportation Act 1995 of Virginia) can be mentioned as examples. The existing framework and amendments vary across countries: thus, for instance, many governments consider the revenues collected in the transport industry as general-purpose taxes, while others place them in sectoral Trust Funds and use them for transportation purposes. Nevertheless, the main efforts have been focused everywhere on:

- finding resources other than conventional revenues (such as fuel and vehicle taxes and tolls);
- identifying new roles and responsibilities for the private sector beyond design and construction (up to ownership);
- leveraging existing sources and encouraging private investment.

All these developments do not mean that direct ‘pay-as-you-go’ public financing (based on conventional taxes, combined with public toll management revenues or augmented with other ‘innovative’ local-interest-based contributions) as lost its legitimacy. It remains an applicable model. Toll-collecting authorities work like profit-seeking corporations owned by their home states. Moreover, several local administrations have been able to introduce local option taxes to finance their transportation projects (Beale *et al.*, 1996).

Three basic scenarios of public–private co-operation have been applied, differing basically in the government’s financial involvement and the timing of ultimate ownership take-over:

- BOO—build, own, operate: the facility remains the property of the private consortium for a limited period of time;
- BOT—build, operate, transfer: the ownership of the facility is transferred to the sponsoring government free of charge after a limited time period;
- BTO—build, transfer, operate: the consortium leases the facility from the government and operates it with full financial responsibility.

Sometimes the *design* phase or the *financing* activity is stressed (*DBFO*), and *B* may mean *bid*, as an action of importance. In a risk-reducing BO alternative (preferred in some European countries) no transfer occurs and the users do not pay tolls, but the government pays to the concessionaire according to the traffic counted (*shadow toll*—Nutt and Roden, 1998).

By the end of the 1980s a vast amount of experience had accumulated in various public–private-delivered projects. The promoters of PPP collected, evaluated and formulated the basic conditions, difficulties and principles related to public–private co-operation. The good-faith approach (Ferrigno, 1990; Miller, 2000) takes as proven that private-sector participation results in a combination of lower costs and less risk for

the public sector. It stresses that in a true joint venture, all risks, costs and payments must be shared proportionally. The multiple interests of several participants (e.g. of those expecting returns on their commercial contributions and project equity investments as well) have to be skilfully negotiated and 'packaged'.

Preconditions of a successful infrastructure project are listed in the pro-PPP literature. International institutions, consulting firms and individual experts have identified the general and particular difficulties appearing in the public-private joint developments. Experience shows that there are several basic principles and guidelines worth applying during project preparation. Some have to do with the quality of the participants and the relationships among them. Others more important during the phase when the financing and implementation are negotiated (Scharle, 2001).

Particular attention has been paid to the risks involved in PPP projects. On the one hand, private entities are aware of their risks and expect commensurate rewards for taking them. On the other hand, both governments and concessionaires have political and financial motives to pursue projects and conclude agreements without assessing risks and their corresponding rewards properly. Risks may occur during any phase of project delivery. Their assessment is in the interests of all the players. The principal risks concern development, construction (completion) and operation, and are commercial, economic, financial and political (Lockwood, 1997; Fayard, 1998). PPP is an adequate framework for compromising the risks and allocating them to the partner in power to manage them best. The public sector controls several key legal and moral assets (such as land and authority) to implement a project within the context of an overall development programme. The private sector brings outside capital, technical expertise and an incentive structure. The essence is the co-operative and mutually supporting relationship: each party has a stake in the success of the other.

Actual partnering involves close collaboration and the combination of the strengths of both the private sector (more competitive and efficient in economic terms) and the public sector (more responsible and accountable to society). It is commonly acknowledged that a greater involvement of the private sector reveals problems inherent in public institutions, activities and procedures. PPP can bring the discipline of the market into public administration.

The PPP financing approach highlights the utmost importance of identifying the parties interested in the implementation of a project. Different groups of users (such as residents, commuters, hauliers, tourists, etc.), mobility-businesses (such as food, lodging, fuel supply, etc.) and other entities (such as property owners, developers, vendors, etc.) can be assessed as beneficiaries. All can be invited to contribute to the implementation costs according to the improvements they are given or expect to enjoy. An appropriate segregation of the 'general public' is both possible and necessary in order to allocate the burdens fairly (Beale *et al.*, 1996). Right-of-way donations, abutter impact fees, exactions, value-capture taxes or development taxes can be combined with exemptions and waivers and increase the resources drawn from traditional taxes (of head, sales, income, property, etc.) and direct tolls. Bonds, loans, and other financing techniques can be applied to bridge, distribute and phase the burdens among the parties during the whole project lifetime.

Experts and professionals having no direct engagement on the private side are less enthusiastic about PPP. Senior executives with long service time in different governmental positions question both the all-round merits of the private partner and the general culpability of public administration performance. Academic observers cast doubt on the innovative character of the partnership. They stress that private ownership and the

operation of roads and other transportation facilities have been common in different cultures and ages and that only the mass mobilization of the twentieth century forced the public authorities to take responsibility for constructing, operating and maintaining the road transportation infrastructure. Sociologists highlight the secondary effects of PPP schemes and a few of them regard commercialization as an adequate answer only when the efficiency of the administration is discussed.

Many observers appreciate that the one feature all PPPs have in common is a shared responsibility for infrastructure policy and its implementation, as this affects citizens (Rosenau, 1999). In road transport infrastructure PPPs, for instance, both partners want to see as many users as possible, but for different reasons. The respective roles of the private and the public partner are neither antagonistic nor identical but complementary: the government wants to improve a service, to stimulate economic growth in the region and to ensure a fair return on everybody's investment; the private partner wants to maximize its income at the highest possible specific revenues.

Promoters of PPP cautiously avoid blaming the public. They focus their criticism on governments. Unbiased observers identify weaknesses in the attitude of the general public towards transportation infrastructure financing and argue that the public is seldom aware that:

- no institution is able to spread the expenses of the construction, operation and maintenance of the road network so that all individuals or groups of society are charged or taxed in proportion to accrued benefits;
- though public administrations are politically neutral organizations serving long-term public interests, they work under the influence of short-term political pressures;
- governments are often given new tasks while it is difficult for them to abandon outdated activities;
- the disparity between resources and costs is independent of the techniques used to collect and redistribute public revenues. Both the Highway Trust Fund-type financing and the budget-based fiscal one may suffer from the same problem: revenues fall below the reasonable asset maintenance burden.

Academic and other unbiased experts acknowledge the main features of private efficiency stressed by the advocates of PPP (such as better management, cost reduction, greater operating efficiency, etc.—Morgan, pers. commun.). They disclose further facts such as:

- managers in the private sector are paid to account for and to manage risks while in the public sector they are paid to avoid them;
- public agencies are (often tacitly) expected to provide side-services (such as jobs and social benefits for handicapped people) without clear financing backgrounds;
- public agencies are predestined to invest in facilities with long-term positive externalities while private entities invest in profit-making ventures;
- the accounting profession often reduces the capital costs of the private partner to a level that a public agency cannot match in the given legal environment.

Advocates of PPP frequently state (or, occasionally, fail to deny) that innovative financing generates more money and mobilizes new resources. Senior public servants stress that the basic antagonism existing between the heavy taxation of the transportation industry and the chronic cutback of budget spending on transportation infrastructure cannot be resolved by any institutional restructuring. Some observers are open and brave enough to point out that innovative financing does not create any revenues at all (Mineta,³ 1997). Revenues come from the taxpayers (charged with general or specific taxes) and from the

beneficiaries (direct or indirect ones including road users, local communities, developers and others favoured by the public facility—Fayard, 1998).

Case study analyses prove that local authorities, residents and private companies involved in small-scale real estate developments can easily rearrange their roles and rules in the road improvement scenario. Responsibilities, costs (e.g. of constructing and operating), and rights (e.g. of taxing and charging) can be clearly defined, bargained and balanced at this level. The basic question as to ‘who is paying for what’ is simple to answer. This identification does become difficult or even impossible when the benefits and beneficiaries of the road network as a whole are considered at the regional or national level.

So far, very few PPP failures have been reported openly. Neither the public nor the private partner is interested in disclosing data or facts; outsiders with responsibility do not enter into fictitious analyses. A few of the examples show, however, that serious problems can occur even in ostensibly regular cases (Bloomfield *et al.*, 1998). PPP does not mean ‘less government’ but a different governmental role. Because of the stronger private positions, more skilled government participation is needed. Public administration must not ‘let the private sector excessively use the public credit for private gain’; it ‘cannot have the private sector cherry-pick projects’ (Giglio, 1997).

Professional skill and business-like operation are not equal to business itself. The mechanical (rather than organic) managerial reorganization of public services may result in unwelcome, unexpected and irreversible side-effects. Experience gained in New Zealand, a small country with traditional egalitarianism, where one of the most radical and coherent sets of privatization measures was introduced to separate high-level authorities from their trading organizations, shows that:

- a technocratic implementation destroyed the traditional public service career mechanism;
- interdependence between public entities shifted to independence, while collectivist norms shifted to individualistic norms;
- reductionist measures instilled competitive norms instead of co-operative ones;
- contractual outsourcing increased the risk of corrupt practice at the interface between private business and governmental discretion;
- contractual appointments opened up the public service at senior levels for people with little understanding of the differences between public and corporate management (Gregory, 1999).

Independent observers send significant messages to both partners:

- if it was contracted fairly, PPP may prove to be a low-margin business for the private sector;
- there are no cast-iron laws governing public infrastructure financing overtime. Tax-based and toll-based financing can coexist independently from PPP schemes;
- a remarkable number of difficulties experienced in PPP cases occurs as a consequence of the partnership itself. They do not appear or prove to be negligible when conventional pay-as-you-go contracting is applied;
- some of the ‘weaknesses’ of the partners (such as bureaucratic or monetary risk aversion) are of an institutional origin and have significant positive functions in regular cases;
- the commercialization of public services cannot be separated from the ethical environment in a given society. Moral standards cannot be maintained when institutional changes diminish social capital.

Perspectives offered by other disciplines

Linguistics

In being saturated with political and social intentions and motivations, PPP players are inclined to camouflage some arguments or use ambiguous terms for convenience. The risk of confusion is not specific to PPP but is higher than usual. A recent classification (Fayard, 1998) and professional linguistic analysis (Linder, 1999) helps us understand some ideological aspects reflected in the promotional debates about PPP.

The term *partnership* has multiple meanings connected with the two main ideological streams of neo-liberalism and neo-conservatism:

- neo-liberalism assumes that moral regeneration can be achieved through ownership and market exchange; market incentives foster spontaneous social co-ordination; the key to progress is not community solidarity but the set of novel organizational mechanisms;
- neo-conservatism assumes that self-reliance and commerce (both based on consolidated traditional knowledge) guarantee human freedom; local cultural institutions (such as family, community churches and civic associations) inculcate the essential values undermined by concentrated economic or political power.

In these ideologies two contradictory claims of espousing traditional moral and free market values are lumped together. Both believe that the individual can be both citizen and consumer; social life can be divided into separate domains: a private one for money and consumption and a public one for power and citizenship. Both ideologies reconsider the role of the state as an antithesis of private entities: neo-liberals presume that state transactions are inefficient; neo-conservatives think it is overburdened and under-disciplined.

Based on these considerations, at least six meanings of partnership can be distinguished in the promotional, political and professional literature (Linder, 1999):

- *Management reform*—skills to find new markets, to enhance productivity and to stay ahead of the competition are needed to improve government performance. Government managers must become more occupied with making deals than with administrative procedures. This meaning is prototypically neo-liberal.
- *Problem conversion*—government managers have to commercialize problems. They act as an investment broker, attracting entrepreneurs and private capital to co-operate on a project-by-project basis. This neo-liberal interpretation implies that government managers promote private business growth.
- *Moral regeneration*—partnerships are intended to have a moral effect on the people involved. Having given the public a direct stake in an important community enterprise accomplishes the moral purpose of giving people a chance for ownership better than shares of utilities sold by the state on the stock market. Neo-liberals find virtues in the market; neo-conservatives think the market offers the expression of certain values learnt elsewhere.
- *Risk shifting*—risk shifting and problem conversion differ more in degree than in kind. However, the idea of heightened vigilance over public investments squares with neo-conservative efforts to restrain government spending. Risk shifting assigns the supporting role to commercial interests.
- *Restructuring public service*—partnerships can move from a public to a private workforce disciplined by the labour market and can serve as a means of deregulating employ-